Arduino Uno Voltage Protector

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Objective

The aim of this project is to disconnect a load when the supply voltage or current is outside a safe range. I used an Arduino Uno, two potentiometers, an I^2C LCD and four buttons. The potentiometers represent voltage and current. A green LED represents the connected load, and a red LED shows a fault. All electrical signals in the simulation are low voltage.

How the system works

The Arduino reads the voltage input at A0 and the current input at A1. Their ranges represent 0–300 V and 0–20 A. The default limits are shown below. If voltage is below or above its limit, current is too high, or an ADC input is saturated, the load output at D8 goes LOW. The first fault stays latched, even after the input returns to normal.

Minimum voltage	Maximum voltage	Current limit	Restart delay
180 V	250 V	5 A	3 s

RESET is accepted only when the voltage is at least 5 V inside both limits and current is at least 0.5 A below its limit. The inputs must then stay safe for the full restart delay. This prevents repeated switching near a limit. The load also starts OFF when the Arduino boots.

Settings and communication

MENU selects a setting; PLUS and MINUS change it. The LCD shows the measured values, output state and selected setting. The I^2C display uses A4 (SDA) and A5 (SCL), which reduces circuit wiring. The computer uses USB serial at 115200 baud. For example, `SET VMAX 245` sets the upper voltage limit, `SET IMAX 4000` sets a 4 A limit, and `GET` reads the status. Both local and serial changes use the same validation. Invalid values are rejected. Settings are held in RAM and return to defaults after restart.

Testing and results

The firmware compiled successfully for Arduino Uno. All 37 protection tests and 19 integration checks passed using compiled AVR code. They cover startup delay, voltage and current faults, safe reset, hysteresis, timer rollover, buttons and serial commands. Wokwi also demonstrated normal operation, fault latching, local settings and computer settings. In two emulator injections, D8 went LOW after 3.982 ms for over-voltage and 0.878 ms for over-current. Screenshots, serial logs, CSV events, VCD capture and timing plots are included in `test_evidence/`. These are simulation results, not physical measurements.

Limitations

This version uses DC input levels and an LED load. It does not measure real AC RMS voltage or switch an appliance. The LCD is refreshed one character per loop so protection checks continue between characters. Sampling is nominally every 5 ms; the observed timing is not a worst-case guarantee, and very short surges may be missed. A physical version would need suitable sensing and switching hardware and separate timing tests. No mains circuit was built.

Project files: Arduino source, Wokwi simulation, wiring diagram and test evidence.